

+ **MEDI-CARE** V1.0 SYSTEM

Pharmacy Management System

| PROJECT SNAPSHOT



Application UI

VB.NET Windows Forms application designed for streamlined pharmacy operations.



Database Architecture

Powered by Microsoft Access database with seamless OleDb connectivity.



Core Modules

Integrated customer, employee, stock inventory, billing, and sales-history modules.

1. Problem Statement & Objectives



Problem Statement

Challenges with traditional pharmacy management

⊖ Manual Management Inefficiencies

Manual pharmacy work makes customer records, stock tracking, and billing harder to manage accurately.

⊖ Fragmented Operations

Lack of a centralized desktop application to unify routine daily pharmacy tasks.

⊖ Redundant Data Entry

Repetitive data entry causes delays and errors without seamless integration between stock, customers, and billing.



Key Objectives

Core goals designed to automate and streamline workflow

01

Information Management

Maintain complete customer and employee information in a centralized database.

02

Stock & Item Tracking

Maintain medicine and item stock levels along with rates and expiry dates.

03

Automated Billing & Calculations

Create bills with instant calculation of GST, discounts, and gross total.

04

Real-Time Inventory & Print Retrieval

Automatically reduce stock after billing, store details for retrieval, and provide a menu-driven Windows interface.

2. Technology Stack



VB.NET

Application language

Visual Basic .NET



Windows Forms

UI framework

Desktop forms + DataGridView



MS Access

Database

medicare.accdb



OleDb

Connectivity

System.Data.OleDb



.NET Framework

Runtime

Target: 4.0 Client



Printing

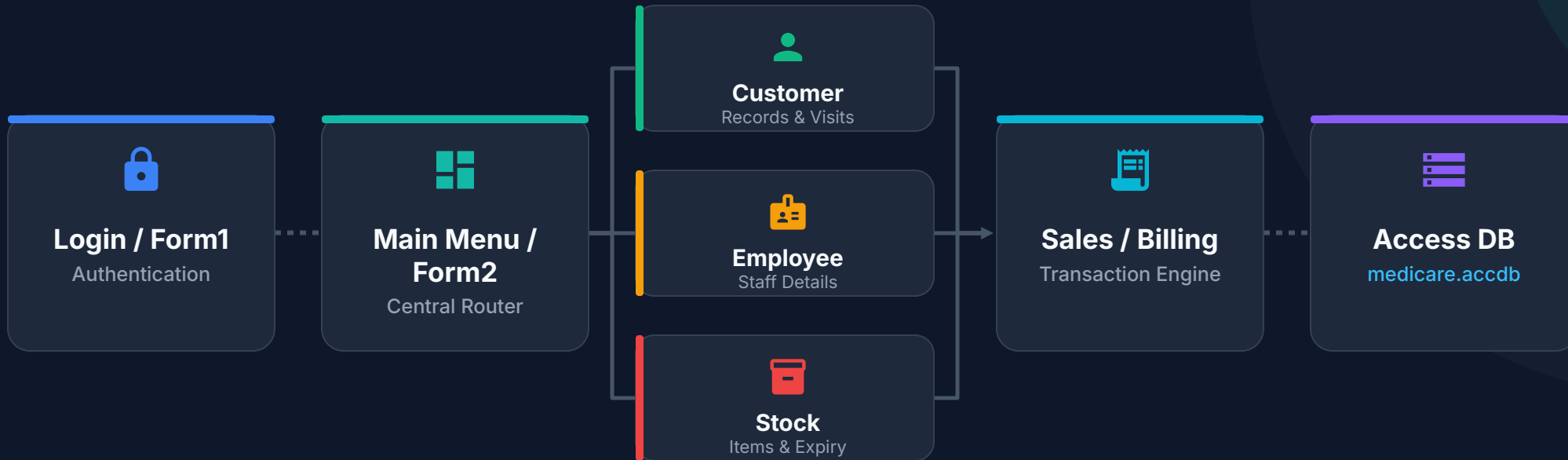
Output

PrintDocument + PrintPreviewDialog



Build target: x86 / WinExe

3. System Architecture & Data Flow



CORE DATA PATTERN

01 Form Events → 02 OleDbCommand / DataAdapter → 03 Access Tables → 04 DataGridView / Billing Output

4. Major Modules



Authentication

Form: Form1

Admin login + password
visibility toggle



Main menu

Form: Form2

Sales, Customer, Employee,
Records, Stock



Customer

Form: customer

CRUD + visit count



Employee

Form: employee

CRUD + employee details



Stock

Form: stock

CRUD + item/price/expiry data



Sales

Form: sales

Bill creation, calculations, stock
update



Records

Form: record

Counts of employees,
customers and items



Bill details

Form: billdeatil

Retrieve bill by date/number +
print preview

5. Login & Navigation



Login Implementation

Form: Form1

- **Authentication Check:** Form1 validates entered username and password against records.
- **Default Credentials:** username = bala password = guru
- **Visibility Toggle:** Dynamic checkbox toggles password character masking.
- **Form Transition:** Valid authentication launches Form2 and hides Form1.

💡 Viva point: event-driven Windows Forms



Navigation

Form: Form2

- **Main Hub:** Form2 acts as the main module router and selector.
- **Sales Workspace:** Sales selection opens the dedicated MDI workspace.
- **Sub-Modules:** Customer, Employee, Stock, and Records launch respective forms.
- **MDI Switching:** MDI menu dynamically switches between Sales Entry and Sales Details.

💡 Viva point: MDI = Multiple Document Interface

6. Master Data: Customer, Employee & Stock



Customer

Form: customer

Fields: ID, name, address, first visit date, visit count.

CRUD Operations:

- **SAVE** → INSERT
- **UPDATE** → UPDATE
- **DELETE** → DELETE

DataGridView reloads records automatically on updates.



Employee

Form: employee

Fields: ID, name, phone, email, address, salary, joining date.

- **Data Engine:** CRUD operations are handled through **OleDbCommand**.
- **Grid View:** DataGridView displays the full employee table.
- **Validation:** KeyPress handlers restrict input on numeric fields.



Stock

Form: stock

Fields: Item ID/name, quantity, item rate, sell rate.

- **Tracking:** Item entry, manufacture and expiry dates.
- **Operations:** CRUD operations with numeric input validation.
- **Form Sync:** Selected grid row is automatically copied back into form fields.

7. Sales & Billing Workflow

STEP 01 

Enter Customer ID

Fetches customer name + visit count

STEP 02 

Select Item

Loads sell rate, available quantity and item code

STEP 03 

Enter Quantity

Blocks quantity above available stock

STEP 04 

Add Item

Adds line to DataGridView and recalculates totals

STEP 05 

Generate Bill

Stores purchase header + bill line items

STEP 06 

Update Data

Increments visits and decreases stock

STEP 07 

Print

PrintPreviewDialog shows receipt

COMPLETE FLOW 

Transaction Output

💡 Viva point: End-to-end automated POS workflow with database sync

8. Billing Calculations (as implemented in the project)



Calculation Logic

Base Formula

TOTAL = sum of line-item amounts

GST = TOTAL × 0.28 (28%)

Discount Condition

If customer visits > 4:

DISCOUNT = (TOTAL + GST) × 0.04

Otherwise: DISCOUNT = 0

Final Payable

GROSS TOTAL = TOTAL + GST - DISCOUNT



Example Walkthrough

Line-Items Total ₹1,000

GST (28%) + ₹280

Customer Visits: 5 (> 4 → Discount Applies)

Discount (4% of ₹1,280) - ₹51.20

Gross Total Payable

₹1,000 + ₹280 - ₹51.20

₹1,228.80



Important: these percentages are project-specific logic

SYSTEM ARCHITECTURE

customer

Schema Attributes

cid (PK), cname, caddress, fdate, no_visi

Stores customer profile and visitation counters for discount evaluation.

employee

Schema Attributes

eid (PK), ename, ephone, email, eaddress, esalary, jdate

Manages staff records, contact info, salaries, and join dates.

stock

Schema Attributes

itemid (PK), itemname, itemquan, itemrate, sellrate, idate, mdate, xdate

Tracks medicine inventory, pricing, quantity, and expiry dates.

purchase

Schema Attributes

bno (PK), bdate, **cid** (FK), total, gst, discount, gtotal, bpath

Maintains invoice master records with tax and discount totals.

bill

Schema Attributes

bdate, **bno** (FK), **cid** (FK), cname, itemno, itemname, itemquan, itemrate

Stores granular line items for every billed purchase.

Database Configuration

Database File

medicare.acddb

OleDb Provider

Microsoft.ACE.OleDb.12.0

10. Bill Storage, History & Printing



When a Bill is Saved

Database Operations

purchase table stores bill-level metadata and overall totals.

bill table stores granular individual item line records.

customer.no_visi visitation counter is automatically incremented.

stock.itemquan stock quantity is reduced for every item sold.



Sales Details & History

Retrieval Workflow

01. Choose a specific transaction date
02. Select target bill number
03. Fetch customer details + visit count
04. Load all associated bill line items
05. Recalculate and display bill totals



Printing System

Print Integration

Powered by .NET **PrintDocument** & **PrintPreviewDialog** controls.

Receipt Breakdown

Includes shop name, invoice ID, customer info, itemized list, total, GST, discount & net payable.

Custom Page Setup

A dedicated custom paper size is pre-configured for physical thermal receipts.

11. Strengths, Limitations & Future Scope



Strengths

- Simple GUI suitable for desktop use.
- CRUD operations are implemented.
- Stock validation prevents over-selling.
- Billing and receipt printing are integrated.
- Sales history can be retrieved.



Current Limitations

- Login credentials are hard-coded in Form1.
- Several forms use a hard-coded local database path.
- SQL strings are concatenated instead of parameterized.
- No role-based access control is implemented.
- The project targets an older .NET Framework / desktop stack.



Future Improvements

- Use parameterized queries.
- Move credentials to a secure user table.
- Use a configurable connection string everywhere.
- Add expiry alerts and low-stock alerts.
- Add reports/dashboard and role-based permissions.
- Migrate to a server DB for multi-user deployment.

12. Most Likely Viva Questions — Quick Answers

Why VB.NET?

It provides event-driven Windows Forms development with straightforward UI and database integration.

What is OleDb?

ADO.NET provider technology used here to connect VB.NET to the Access .accdB database.

How is stock controlled?

Requested quantity is compared with available quantity; after billing, stock quantity is reduced.

Why DataGridView?

It provides a tabular view for database records and bill line items.

How is printing done?

PrintDocument draws the receipt and PrintPreviewDialog previews it before printing.

Why MS Access?

It is lightweight and suitable for a small single-desktop pharmacy application.

What is CRUD?

Create, Read, Update and Delete — used in customer, employee and stock modules.

How is the bill calculated?

The project sums line totals, applies 28% GST, conditionally applies 4% discount, then computes gross total.

What is MDI?

Multiple Document Interface; the project uses it to host sales-related child forms.

What would you improve first?

Security and reliability: parameterized SQL, secure authentication and configurable DB path.

13. Suggested Live Demo Order



1. Login

bala / guru

Authenticate system user to access main portal.



2. Main Menu

Show Sales, Customer, Employee, Records & Stock navigation links.



3. Stock

Add or show medicine items and verify available quantities.



4. Customer

Display customer records and historical visit counters.



5. Sales

Select customer → item → enter quantity → click ADD.



6. Billing

Demonstrate GST calculation, discount logic, and gross total.



7. Save Bill

Show automated stock reduction and visit count increment.



8. Sales Details

Retrieve bill record and trigger Print Preview display.

KEY TAKEAWAY

"Medi-Care automates the core pharmacy workflow from master-data management to billing and stock updates."